

2.重载和多态

最大值输出

C++

```
1 //定义重载函数maxinum，分别求三个整数、三个浮点数和三个字符中最大的一个并输出。
2 //输入样例1: 45 67 100 34 2.3 78.2 a A B
3 //输出样例1: 100 78.2 a
4 //输入样例2: 0 1 2 2.3 3.4 8.0 d r z
5 //输出样例2: 2 8.0 z
6 #include <iostream>
7 using namespace std;
8 int maxinum(int b1,int b2,int b3){
9     int max=b1;
10    if (b2>max){
11        max=b2;}
12    if (b3>max){
13        max=b3;}
14    return max;
15 }
16 float maxinum(float c1,float c2,float c3){
17     float max=c1;
18     if (c2>max){
19         max=c2;}
20     if (c3>max){
21         max=c3;}
22     return max;;
23 }
24 char maxinum(char a1,char a2,char a3){
25     int max=a1;
26     if (a2>max){
27         max=a2;}
28     if (a3>max){
29         max=a3;}
30     return max;
31 }
32 //StudybarCommentBegin
33 int main()
34 {
35     int b1, b2, b3;
36     float c1, c2, c3;
37     char a1, a2, a3;
38     cin >> b1 >> b2 >> b3 >> c1 >> c2 >> c3>> a1 >> a2 >> a3;
39     cout << maxinum(b1, b2, b3) << ' ' << maxinum(c1, c2, c3) << ' ' << maxinum(a1, a2, a3);
40     return 0;
41 }
42 //StudybarCommentEnd
```

中间值

C++

```
1 #include<iostream>
2 using namespace std;
3 int middle(int arr[]) {
4     int middle1 = arr[0];
5     if ((arr[1]> middle1 && arr[2] > middle1) || (arr[1] < middle1 && arr[2] < middle1)) {
6         middle1 = arr[1];
7     }
8     if (arr[2] < middle1) {
9         middle1 = arr[2];
10    }
```

```
11     return middle1;}
12
13     double middle(double arr[]) {
14         double middle1 = arr[0];
15         if ((arr[1]> middle1 && arr[2] > middle1) || (arr[1] < middle1 && arr[2] < middle1)) {
16             middle1 = arr[1];
17         }
18         if (arr[2] < middle1) {
19             middle1 = arr[2];
20         }
21         return middle1;}
22
23     //在C++中，unsigned 是一种基本数据类型修饰符，用于声明无符号整数类型。无符号整数类型只能表示非负整数（包括 0）。通常情况下，unsigned 用于扩展整数的范围，使其可以表示更大的正整数值，但不能表示负数。
24     //unsigned int x = 10; // 声明一个无符号整型变量 x，赋值为 10
25     //unsigned short y = 65535; // 声明一个无符号短整型变量 y，赋值为最大值 65535
26
27     //在使用 unsigned 时，通常与其他整数类型一起使用，如 unsigned int、unsigned long、unsigned short 等。这些类型声明的变量存储的值都是非负整数。
28
29     unsigned middle(unsigned arr[]) {
30         unsigned middle1 = arr[0];
31         if ((arr[1]> middle1 && arr[2] > middle1) || (arr[1] < middle1 && arr[2] < middle1)) {
32             middle1 = arr[1];
33         }
34         if (arr[2] < middle1) {
35             middle1 = arr[2];
36         }
37         return middle1;}
38
39     //StudybarCommentBegin
40     int main()
41     {
42         int iNumber[3]={45,3,89};
43         double dNumber[3]={12.4,6,55.8};
44         unsigned uNumber[3]={345,776,565};
45
46         cout<<"\nThe numbers are ";
47         for( int i=0;i<3;i++)
48             cout<<iNumber[i]<<"\t";
49         cout<<endl<<"the middle of them is "<<middle(iNumber)<<endl;
50
51         cout<<"\nThe numbers are ";
52         for( int i=0;i<3;i++)
53             cout<<dNumber[i]<<"\t";
54         cout<<endl<<"the middle of them is "<<middle(dNumber)<<endl;
55
56         cout<<"\nThe numbers are ";
57         for( int i=0;i<3;i++)
58             cout<<uNumber[i]<<"\t";
59         cout<<endl<<"the middle of them is "<<middle(uNumber)<<endl;
60
61         return 0;
62     }
63     //StudybarCommentEnd
64
65
66
67
```

相加重载

```
1  #include<iostream>
2  using namespace std;
3  int myadd(int a,int b)
4  {
5      return a+b;
6  }
7
8  int myadd(int a,int b,int c)
9  {
10     return a+b+c;
11 }
```

C++

```

12 //StudybarCommentBegin
13 main()
14 {
15 cout<<myadd(2,3)<<"\n";
16 cout<<myadd(2,3,4);
17 }
18
19 //StudybarCommentEnd

```

数组重载

```

1  #include<iostream>
2  using namespace std;
3  int myadd(int a,int b)
4  {
5      return a+b;
6  }
7
8  int myadd(int a,int b,int c)
9  {
10     return a+b+c;
11 }
12 int myadd(int d[],int n)
13 {
14     int sum =0;
15     for (int i=0;i<n;i++){
16         sum=sum+d[i];
17     }
18     return sum;
19 }
20 //StudybarCommentBegin
21 main()
22 {
23     int a[20],n,i;
24
25     cin>>n;
26     if(n==2)
27         cout<<myadd(5,3);
28     else if (n==3)
29         cout<<myadd(5,3,9);
30     else if(n>3)
31     {
32         for(i=0;i<n;i++)
33             {cin>>a[i];}
34         cout<<myadd(a,n);}}
35 //StudybarCommentEnd

```

C++

时间类重载

```

1  #include<iostream>
2  using namespace std;
3  class Cmytime{
4      private:
5          int h,m,s;
6      public:
7          Cmytime() : h(0), m(0), s(0) {} // Default constructor
8          Cmytime(int hh) : h(hh), m(0), s(0) {}
9          Cmytime(int hh, int mm, int ss) : h(hh), m(mm), s(ss) {} // Constructor with parameters
10         int Set(int hh,int mm,int ss){
11             h=hh;
12             m=mm;
13             s=ss;
14             return 0;}
15         void Show() {
16             cout << h << ":" << m << ":" << s;
17         }
18     };

```

C++

```

18 //StudybarCommentBegin
19 int main(void) {
20     int h,m,s;
21     cin>>h>>m>>s;
22     Cmytime t1(3,2,1),t2,t3(5);
23     t1.Show();
24     cout<<"\n";
25     t1.Set(h,m,s);
26     t1.Show();
27     cout<<"\n";
28     t2.Show();
29     cout<<"\n";
30     t3.Show();
31     return 0;}
32 //StudybarCommentEnd
33

```

时间重载 2

```

1 //设置Cmytime类。
2 //
3 //具有三个成员函数
4 //
5 //构造函数
6 //
7 //Set(int h,int m,int s)
8 //
9 //Show()
10 //
11 //输入  23  25 38
12 //
13 //输出:
14 //
15 //3:2:1
16 //23:25:38
17 //0:0:0
18 //5:0:0
19 #include<iostream>
20 using namespace std;
21 class Cmytime{
22     private:
23         int h,m,s;
24     public:
25         Cmytime() : h(0), m(0), s(0) {} // Default constructor
26         Cmytime(int hh) : h(hh), m(0), s(0) {}
27         Cmytime(int hh, int mm, int ss) : h(hh), m(mm), s(ss) {} // 含参构造函数
28         int Set(int hh,int mm,int ss){
29             h=hh;
30             m=mm;
31             s=ss;
32             return 0;}
33         void Show() {
34             cout << h << ":" << m << ":" << s;}};
35 //StudybarCommentBegin
36 int main(void) {
37     int h,m,s;
38     cin>>h>>m>>s;
39     Cmytime t1(3,2,1),t2,t3(5);
40     t1.Show();
41     cout<<"\n";
42     t1.Set(h,m,s);
43     t1.Show();
44     cout<<"\n";
45     t2.Show();
46     cout<<"\n";
47     t3.Show();
48     return 0;}
49 //StudybarCommentEnd

```

C++

复数类输出

C++

```
1  #include<iostream>
2  using namespace std;
3  class Cmycomplex{
4  private:
5      int num1;
6      int num2;
7  public:
8      Cmycomplex(int n1, int n2) {
9          num1 = n1;
10         num2 = n2;}
11     Cmycomplex() {
12         num1 = 0;
13         num2 = 0;}
14     Cmycomplex(int n1) {
15         num1 = n1;
16         num2 = 0;}
17     void Show() {
18         cout <<"(" << num1 << "+" << num2<<"i" << ")" << endl;};};
19 //StudybarCommentBegin
20 int main()
21 {Cmycomplex  z1(2,3),z2,z3(3);
22 z1.Show();}
23 //StudybarCommentEnd
```

复数输出实部和虚部

C++

```
1  //如题，要求实现：
2  //1、复数类含两个参数的构造函数，一为实部，一个为虚部
3  //2、用GetReal() GetImaginary()返回实部、虚部，均为double型
4  #include<iostream>
5  using namespace std;
6  class Cmycomplex{
7  private:
8      int num1;
9      int num2;
10 public:
11     Cmycomplex(int n1, int n2) {
12         num1 = n1;
13         num2 = n2;}
14     Cmycomplex() {
15         num1 = 0;
16         num2 = 0; }
17     Cmycomplex(int n1) {
18         num1 = n1;
19         num2 = 0;}
20     int GetReal(){
21         return num1;
22     }
23     int GetImaginary(){
24         return num2;}
25     void Show() {
26         cout <<"(" << num1 << "+" << num2<<"i" << ")" << endl;};};
27 //StudybarCommentBegin
28 int main()
29 {Cmycomplex  z1(2,3),z2,z3(3);
30 cout<<z1.GetReal()<<"\n";
31 cout<<z2.GetImaginary()<<"\n";
32 cout<<z3.GetReal()<<"\n";
33 }
34 //StudybarCommentEnd
```

C++ ▾

复制

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